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Master's thesis

Counting lines

Any short subtitle

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Contents

Chapter 1

Introduction

intro

1.1 Notation and important results

Start with some notation and conventions that will be used throughout the thesis.

We will now present some important results that will be used later in the thesis.

thm:bertini

Theorem 1.1.1 (Bertini). *If $\mathcal{D}$ is a linear system on a variety X in characteristic 0, then the general member of $\mathcal{D}$ is smooth away from the base locus of $\mathcal{D}$ and the singular locus of X.*

Add notation (this is from EH): A *linear system* on a scheme X is a pair $\mathcal{D} = (\mathcal{L}, V)$, where $\mathcal{L}$ is a line bundle on X and $V \subseteq H^0(X, \mathcal{L})$ a vector space of sections.

We refer to (!?) for a proof. [EH16]

Find a good reference for this theorem.

thm:bezout

Theorem 1.1.2 (Bézout). *Let $X, Y \subseteq \mathbb{P}^n$ be two closed subvarieties having complementary dimension, i.e. $\dim(X) + \dim(Y) = n$. If X and Y intersect transversely, then they will intersect in exactly $\deg(X) \cdot \deg(Y)$ points.*

We refer to (!?) for a proof. [EH16] corollary 2.4.

Find a good reference for this theorem. Don't require transverse intersections.

Jørgen: Is X and Y intersect in finitely many points, then the number of intersection points is at most deg(x)deg(Y).

thm:fiber

Theorem 1.1.3 (Dimension formula for fibers). *Let k be a field and let $f \colon X \to Y$ be a dominant morphism of irreducible projective varieties over k. Let η denote the generic point of Y and write $X_\eta = X \times_Y \operatorname{Spec} k(Y)$ for the generic fiber. Then*

$$\dim X \;=\; \dim Y \;+\; \dim X_\eta.$$

Write the proof, and/or find reference. Maybe use Vakil 12.4.2.

1.2 Notation

We will use the following notation throughout the thesis:

- $K(X)$ denotes the function field of a variety X.
- $K(X)^\times$ denotes the multiplicative group of non-zero elements in $K(X)$.
- $\Bbbk$ denotes an algebraically closed field of characteristic 0.

1.3 Blow-ups, and the exceptional divisor

Decide if this is viable/necessary for the thesis. Technically use this in the 27 lines proof, but could assume this is known, and just give blow ups in relation to divisors.

Chapter 2

Grassmannians and the space of lines in $\mathbb{P}^3$

assmannians

In this chapter we will introduce Grassmann varieties, or Grassmannians, and focus on the space of lines in $\mathbb{P}^3$, which is denoted $\mathbb{G}(1,3)$. We will also describe the Grassmannians as projective by introducing its affine cover, and a way to embed the Grassmannian into a projective space, giving it the structure of a projective variety. For a more detailed introduction to Grassmannians, we refer to [EH16] Chapter 3 and 4, or [EO25] section 24.6. We closely follow the exposition in [EO25]. (!?)

2.1 Definition and basic properties

Remark 2.1.1. We will use k to refer to the dimension of a linear subspace, and $\Bbbk$ to refer to a field.

A Grassmannian is a projective variety whose closed points correspond to linear subspaces of a certain dimension in a vector space. As a set, $\mathrm{Gr}(k, V)$ is the set of all k-dimensional linear subspaces of the n-dimensional vector space V. In our case, V will be a vector space over an algebraically closed field $\Bbbk$.

Definition 2.1.2. [EO25] We define the *Grassmannian* $\mathrm{Gr}(k, V)$ of k-dimensional linear subspaces of V as the set

$$\mathrm{Gr}(k,n)(\Bbbk) = \{[L] \mid L \subseteq \Bbbk^n \text{ is a linear space of dimension } k\}.$$

A k-dimensional linear subspace of an n-dimensional vector space V can also be thought of as a $(k-1)$-dimensional linear subspace of $\mathbb{P}V \cong \mathbb{P}^{n-1}$. Due to this, we will often write $\mathrm{Gr}(k, V)$ as $\mathbb{G}(k-1, \mathbb{P}V)$. When there is no need to specify the vector space V, only its dimension n, we will write $\mathrm{Gr}(k,n) = \mathbb{G}(k-1, n-1)$. In this thesis we will most often write $\mathbb{G}(k-1, n-1)$, as it will be more natural when working with linear subspaces of projective space. For example, $\mathbb{G}(1,3)$ parametrizes lines in $\mathbb{P}^3$.

The Grassmannian is an important and very useful structure in algebraic geometry. Naturally, when asking questions regarding lines, planes, or other linear subspaces of projective spaces, many answers include the geometry of this variety of the linear subspaces themselves.

Not discussed in this thesis is the theory of Schubert calculus.

Continue discussion.

2.2 Covering by affine spaces

- Idea of covering by affine spaces, referencing EO and EH.
- Example: $\mathbb{G}(1,3)$.

To describe the Grassmannian as a projective variety, we will describe an affine cover of the Grassmannian. Section 24.6 of [EO25] goes into detail about this construction, shows that the cover glues to a scheme. We will go over the overall idea, but exclude the finer details.

We represent a linear subspace $L \subseteq \mathbb{k}^n$ as the kernel of a surjective linear map $\mathbb{k}^n \to \mathbb{k}^{n-k}$. This map can be represented by an $(n-k)\times n$ matrix A of rank $(n-k)$. We have $L = \operatorname{Ker}(A) \subseteq \mathbb{k}^n$, where

$$A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{(n-k),1} & \cdots & a_{(n-k),n} \end{pmatrix}.$$

Note that the matrix A is not unique for a given linear subspace L, since multiplying A on the left by an invertible $(n-k)\times(n-k)$ matrix $G \in \mathrm{GL}_{n-k}(\mathbb{k})$ gives another matrix GA with the same kernel as A. For this reason, we consider the equivalence relation on the set of $(n-k)\times n$ matrices of rank $(n-k)$ given by $A \sim A'$ if there exists an invertible $(n-k)\times(n-k)$ matrix G such that $A' = GA$. The Grassmannian $\mathrm{Gr}(k,n)$ can then be identified with the set of equivalence classes of $(n-k)\times n$ matrices of rank $(n-k)$ under this equivalence relation. We notice that if L is represented by a matrix A, we can find a more canonical example by instead regarding A in its reduced echelon form. In this way, we represent L by a matrix A with some $(n-k)\times(n-k)$ identity matrix as a submatrix. At the same time, a matrix A determines a subspace $L \subseteq \mathbb{k}^n$, and two matrices in reduced echelon form represent the same subspace if and only if they are equal. Note that a matrix in reduced echelon form with an $(n-k)\times(n-k)$ identity matrix as a submatrix is parametrized by an affine space of dimension $(n-k)k$.

Example 2.2.1. The matrices

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 2 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

since they represent the same 2-dimensional subspace of $\mathbb{k}$, and is in the same class as the matrix their reduced echelon form

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

▲

For these reasons, we consider the set

$$(\mathbb{A}^{(n-k)n}(\mathbb{k}) - Z)/\mathrm{GL}_{n-k}(\mathbb{k})$$

where $\mathbb{A}^{(n-k)n}(\mathbb{k})$ parametrizes the set of all $(n-k)\times n$ matrices with entries in $\mathbb{k}$, and Z is the closed subset of $(n-k)\times n$ matrices of rank less than $(n-k)$. The $\mathbb{k}$-points of $\mathrm{Gr}(k,n)$ is bijective to the classes in this set. We now define an open cover of $\mathrm{Gr}(k,n)$.

Is this correct? How best to describe this?

Let $I \subseteq \{1, \ldots, n\}$ be a subset of size $(n-k)$. For each I let $U_I \subseteq \mathbb{A}^{(n-k)n}(\Bbbk)$ denote the closed subscheme such that the columns given by I form a $(n-k) \times (n-k)$ submatrix which is also the identity matrix. Each U_I is isomorphic to the affine space $\mathbb{A}^{(n-k)k}$. The collection of all such U_I for all subsets I of size $(n-k)$ gives an open cover of $\mathrm{Gr}(k, n)$. Note that there are $\binom{n}{n-k} = \binom{n}{k}$ such subsets, one for each I. [EO25] shows that this cover glues to give the Grassmannian the structure of a scheme.

Example 2.2.2. For $n = 4, k = 2$, there are 6 affine spaces $\mathbb{A}^4$. They are:

$$U_{12} = \begin{pmatrix} 1 & 0 & a_{1,3} & a_{1,4} \\ 0 & 1 & a_{2,3} & a_{2,4} \end{pmatrix} \qquad U_{13} = \begin{pmatrix} 1 & a_{1,2} & 0 & a_{1,4} \\ 0 & a_{2,2} & 1 & a_{2,4} \end{pmatrix}$$
$$U_{14} = \begin{pmatrix} 1 & a_{1,2} & a_{1,3} & 0 \\ 0 & a_{2,2} & a_{2,3} & 1 \end{pmatrix} \qquad U_{23} = \begin{pmatrix} a_{1,1} & 1 & 0 & a_{1,4} \\ a_{2,1} & 0 & 1 & a_{2,4} \end{pmatrix}$$
$$U_{24} = \begin{pmatrix} a_{1,1} & 1 & a_{1,3} & 0 \\ a_{2,1} & 0 & a_{2,3} & 1 \end{pmatrix} \qquad U_{34} = \begin{pmatrix} a_{1,1} & a_{1,2} & 1 & 0 \\ a_{2,1} & a_{2,2} & 0 & 1 \end{pmatrix}$$

▲

Is this precise enough?

One can also show that this construction is possible over all fields by working over the base ring $\mathbb{Z}$. We will not go into details here, but refer to [EO25] section 24.6.

Proposition 2.2.3 ([EO25], Theorem 24.26). *For a field $\Bbbk$, the Grassmannian $\mathrm{Gr}_{\Bbbk}(k, n)$ is a regular projective variety of dimension $k(n-k)$.*

Proof. By construction, the Grassmannian has an affine covering of affine spaces $\mathbb{A}^{(n-k)k}$, so it is regular of dimension $k(n-k)$. The Plücker embedding (see Section 2.3) shows that it is projective.(!?) □

2.3 The Plücker embedding

sec:Plucker

- General idea of Plucker embedding, referencing EO and EH.
- Example: $\mathbb{G}(1, 3)$.

Again we follow [EO25] in describing the Plücker embedding, and keep the exposition brief. We look at $\mathbb{P}^N_{\mathbb{Z}}$ for $N = \binom{n}{k} - 1$. We claim there is a closed embedding

$$\rho \colon \mathrm{Gr}(k, n) \to \mathbb{P}^{\binom{n}{k}-1}.$$

Locally, consider the morphism

$$\rho_I \colon U_I \to \mathbb{P}^{\binom{n}{k}-1}$$

defined by setting for each ring R, and $A \in U_I(R)$

$$\rho_I(A) = [\det(A_J)]_{J \subseteq \{1, \ldots, n\}, |J| = k}.$$

That is, ρ_I sends a matrix to all of its $(n-k) \times (n-k)$ minors.

Lemma 2.3.1 ([EO25], Lemma 24.24). *For each I, the morphism ρ_I is a closed embedding.*

This means that $\mathrm{Gr}(k, n)$ embeds as a closed subscheme of $\mathbb{P}^{\binom{n}{k}-1}$.

2.4 Universal sub and quotient bundles

Fill in. Include prop 3.3 from EH16.

Later, we will need the notion of the universal sub and quotient bundles over a Grassmannian.

[EH16] Let V be an n-dimensional vector space over a field $\Bbbk$. Consider the Grassmannian $\mathrm{Gr}(k, V)$ of k-dimensional linear subspaces of V. Let $\mathcal{V} = \mathrm{Gr}(k, V) \times V$ be the trivial vector bundle over $\mathrm{Gr}(k, V)$ with fiber V.

Definition 2.4.1. The *universal subbundle* $\mathcal{S}$ over $\mathrm{Gr}(k, V)$ is the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$. That is,

$$S_{[L]} = L \subseteq V = \mathcal{V}_{[L]}.$$

The quotient $\mathcal{Q} = \mathcal{V}/\mathcal{S}$ is called the *universal quotient bundle* over $\mathrm{Gr}(k, V)$.

We define the *universal subbundle* $\mathcal{S}$ over $\mathrm{Gr}(k, V)$ as the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$. Similarly, we define the *universal quotient bundle* $\mathcal{Q}$ over $\mathrm{Gr}(k, V)$ as the vector bundle whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the quotient space V/L.

Proposition 2.4.2 ([EH16], Proposition 3.3). *The subset $\mathcal{S}$ of $\mathcal{V}$ whose fiber over a point $[L] \in \mathrm{Gr}(k, V)$ is the subspace $L \subseteq V$ is a vector bundle over $\mathrm{Gr}(k, V)$.*

2.5 Short on Schubert cycles (?)

Decide if we should include or not. Probably not. Is kinda relevant in the proof of 27 lines.

- Consider adding examples from EH16 to illustrate the kinds of questions one can answer after this chapter.

2.6 Incidence correspondence and the universal Fano scheme

We aim to introduce the concept of incidence correspondences (or an incidence variety), which will be a tool we will use several times later in this thesis. We do this by introducing the universal Fano scheme, which is an example of an incidence variety. Generally, an incidence correspondence is a set of pairs (x, y) such that x lies in y, where x and y are elements of X and Y respectively, for some varieties(!?) X and Y. This can be visualized in the following diagram, with projection maps π_1 and π_2:

$$\begin{tikzcd}
& \mathcal{I} = \{(x,y) \in X \times Y \mid x \text{ lies in } y\} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\
X & & Y
\end{tikzcd} \tag{2.1}$$

Now we define an incidence correspondence of k-planes $L \in \mathbb{G}(k, n)$ contained in degree d surfaces in $\mathbb{P}^n$.

Definition 2.6.1. ([EH16] Section 6.1). Let $\mathrm{N} = \binom{n+d}{d} - 1$, so that $\mathbb{P}^N$ parametrizes all degree d hypersurfaces in $\mathbb{P}^n$. The *universal Fano scheme* is defined as

$$\Phi(n, d, k) = \{(X, L) \in \mathbb{P}^N \times \mathbb{G}(k, n) \mid L \subseteq X\}$$

versal Fano

Remark 2.6.2. The *Fano scheme* $F_k(X) \subseteq \mathbb{G}(k, n)$ of a variety $X \subseteq \mathbb{P}^n$ parametrizes k-planes on X. The universal Fano scheme parametrizes k-planes for all varieties of a certain degree in $\mathbb{P}^n$. By taking the fiber over a point $[X] \in \mathbb{P}^N$, that being the class of the surface $X \subseteq \mathbb{P}^n$, we get the Fano scheme $F_k(X)$ for the specified variety. In this sense, taking the fiber over a "general" X results in a "general" Fano scheme $F_k(X)$. Studying the universal Fano scheme can then tell us what we can "expect" when examining a specific Fano scheme. It is therefore a very useful tool in telling us about a general member of the varieties.

Lemma 2.6.3. *$\Phi \subset \mathbb{P}^N \times \mathbb{G}(k, n)$ is a closed subset of $\mathbb{P}^N \times \mathbb{G}(k, n)$.*

Decide if we want to prove, or refer to proof. Can refer to Harris 1995 chapter 6, or perhaps proof of 24.31 in EO

Proposition 2.6.4 ([EH16], Proposition 6.1). *Let $N = \binom{n+d}{d} - 1$. The universal Fano scheme $\Phi = \Phi(n, d, k) \subseteq \mathbb{P}^N \times \mathbb{G}(k, n)$ is a smooth irreducible variety of dimension*

$$\dim \Phi(n, d, k) = N + (k+1)(n-k) - \binom{k+d}{k}.$$

We ask when we expect a general smooth surface X of degree d to contain lines, and remember that this question is answered by looking at the geometry of the Fano scheme $F_k(X)$ for a general smooth surface X. More precisely one asks what dimension the Fano scheme has, therefore deciding what dimension the set of lines in our general surface is. As noted in Remark 2.6.2, $F_k(X)$ can be obtained by taking the fiber over a point $[X] \in \mathbb{P}^N$. We therefore ask what dimension this fiber will have. From Theorem 1.1.3 we get that for a general point $[X]$, $\dim \pi^{-1}([X]) = \dim \Phi - \dim \mathbb{P}^N$. We define this as $\phi(n, d, k) := \dim \Phi - N$. Now we can answer the question of when we expect lines to be contained in our surface by looking at $\phi(n, d, k)$:

EH16
Corollary 2.6.5 ([EH16], corollary 6.2). *(a) The dimension of any component of the family of k-planes on any hypersurface of degree d in $\mathbb{P}^n$ is at least*

$$\phi(n,d,k) := (k+1)(n-k) - \binom{k+d}{k}.$$

(b) If $\phi(n,d,k) < 0$, then the general hypersurface of degree d in $\mathbb{P}^n$ contains no k-planes.

(c) If $\phi(n,d,k) \geq 0$, and the general hypersurface of degree d contains any k-planes, then every hypersurface of degree d contains k-planes, and every component of the family of k-planes on a general hypersurface of degree d has dimension exactly $\phi(n,d,k)$.

Proof.

Decide if we want to prove, or refer to proof.

□

To bring this back to a familiar context of lines on surfaces in $\mathbb{P}^3$, we set $n = 3$ and $k = 1$. We then get that $\phi(3,d,1) = 3 - d$. We use this in the following corollary:

EO25
cor:Fano Corollary 2.6.6. *([EO25], 22.31) For a hypersurface $X \subseteq \mathbb{P}^3$ of degree d, we expect to find no lines on X when $d \geq 4$, finitely many lines when $d = 3$, and positive-dimensional families of lines when $d \leq 2$.*

Decide if we want to prove, or refer to proof.

2.7 The Chow ring

[Ful84] Let X be an algebraic scheme. A *k-cycle* on X is a finite formal sum

$$\Sigma_i m_i[V_i]$$

where the V_i are k-dimensional subvarieties of X, and the m_i are integers. The group of k-cycles on X, denoted $Z_k(X)$, is the free Abelian group generated by the k-cycles. To a k-dimensional subvariety V of X corresponds $[V]$ in $Z_k(X)$. The group of all cycles on X is denoted

$$Z(X) = \bigoplus_k Z_k(X),$$

and its elements are called *cycles*.

Remark 2.7.1. $(n-1)$-cycles are Weil divisors.

For any $(k+1)$-dimensional subvariety W of X, and any non-zero rational function $f \in K(W)^\times$, we can define a k-cycle $[\mathrm{div}(f)]$ on X by

$$[\mathrm{div}(f)] = \sum_V \mathrm{ord}_V(f)[V]$$

where the sum is over all codimension-1 subvarieties V of W. Here, $\mathrm{ord}_V(f)$ is the order function on $K(W)^\times$ defined by the local ring $\mathcal{O}_{V,W}$.

A k-cycle α is *rationally equivalent to zero*, written $\alpha \sim 0$, if there are a finite number of $(k+1)$-dimensional subvarieties W_i of X and $f_i \in K(W_i)^\times$ such that

$$\alpha = \sum_i [\mathrm{div}(f_i)].$$

Since $[\mathrm{div}(f^{-1})] = -[\mathrm{div}(f)]$, the cycles rationally equivalent to zero form a subgroup of $\mathrm{Rat}_k(X) \subseteq Z_k(X)$. The quotient group

$$A_k(X) = Z_k(X)/\mathrm{Rat}_k(X)$$

is called the *k-th Chow group* of X, and its elements are called *Chow classes*. We say that two k-cycles are *rationally equivalent* if they lie in the same class in $A_k(X)$.

Definition 2.7.2. The *Chow group* of X as the direct sum

$$A(X) = \bigoplus_k A_k(X).$$

An element $\alpha \in A^k(X)$ is called a *cycle class*. We say that a cycle is *effective* if all its coefficients are non-negative. A cycle class is *effective* if it can be represented by an effective cycle.

Include example 1.3.1 from Fulton's book?

[EH16] Theorem-definition 1.5 if we want to say that it's a ring.

2.7.1 Alternative definition of rational equivalence

Do we need this definition?

Definition 2.7.3. [EH16] Let $\mathrm{Rat}(X) \subseteq Z(X)$ be the subgroup generated by differences of the form

$$\langle \Phi \cap (\{t_0\} \times X)\rangle - \langle \Phi \cap (\{t_1\} \times X)\rangle$$

where $t_0, t_1 \in \mathbb{P}^1$ and Φ is a subvariety of $\mathbb{P}^1 \times X$ not contained in any fiber $\{t\} \times X$. We say that two cycles are *rationally equivalent* if their difference lies in $\mathrm{Rat}(X)$, and we say that two subschemes are rationally equivalent if their associated cycles are rationally equivalent.

Find a reference showing that this definition is equivalent to the previous one.

Remark 2.7.4. In many cases, it will be natural to study codimension r subvarieties of an n-dimensional variety X, and so we consider the $(n-r)$-th Chow group. We denote this as $A^r(X) = A_{n-r}(X)$. This group is called the *codimension r Chow group* of X.

For more details on the chow ring, see [Ful84] chapter 1, or [EH16] section 1.2.

2.7.2 The moving lemma

Theorem 2.7.5 ([EH16], Theorem 1.6 (The moving lemma)). *Let X be a smooth quasi-projective variety.*

(a) For every $\alpha, \beta \in A(X)$ there are generically transverse cycles $A, B \in Z(X)$ with $[A] = \alpha$ and $[B] = \beta$.

(b) The class $[A \cap B]$ is independent of the choice of such cycles A and B.

2.8 Chern classes

[EH16] Recall that a Cartier divisor is defined through the vanishing loci of sections of line bundles. Chern classes are a way of generalizing this to vector bundles of higher rank, with the first Chern class corresponding to the Cartier divisor case.

Include assumptions on X to say this is an isomorphism.

2.8.1 The first Chern class of a line bundle

([EH16], 1.4.) Let $\mathcal{L}$ be a line bundle on a variety X and σ be a rational section. Then on an open affine set U of a covering of X, we can write $\sigma|_U = f_U/g_U$, and define $\mathrm{Div}(\sigma)|_U = \mathrm{Div}(f_U) - \mathrm{Div}(g_U)$. This definition agrees on the intersections of the open affine sets, and thus defines a Cartier divisor on X. Moreover, if τ is another rational section of $\mathcal{L}$, then $\alpha = \sigma/\tau$ is a well-defined rational function(!?), so $\mathrm{Div}(\sigma)$ and $\mathrm{Div}(\tau)$ are rationally equivalent divisors:

$$\mathrm{Div}(\sigma) - \mathrm{Div}(\tau) = \mathrm{Div}(\alpha) \equiv 0 \bmod \mathrm{Rat}(X).$$

Thus, we can define the first Chern class of the line bundle $\mathcal{L}$ as

$$c_1(\mathcal{L}) = [\mathrm{Div}(\sigma)] \in A^1(X),$$

the rational equivalence class $\mathrm{Div}(\sigma)$ in the Chow group $A^1(X)$, for any nonzero rational section σ. This is well-defined, as it does not depend on the choice of rational section σ.

Example 2.8.1. A consequence is that $c_1(\mathcal{O}_{\mathbb{P}^n}(d))$ is the class of any hypersurface of degree d. ▲

Improve or remove.

[Bea96] Recall that the Picard group $\mathrm{Pic}(X)$ is the group of isomorphism classes of line bundles $\mathcal{L}$ (or of invertible sheaves) on X, with addition given by the tensor product, $[\mathcal{L}] + [\mathcal{L}'] = [\mathcal{L} \otimes \mathcal{L}']$. To every effective divisor D on X, we can associate a line bundle $\mathcal{O}_X(D)$, defined locally as the set of rational functions f on X such that $\mathrm{Div}(f) + D \geq 0$, and a section $s \in H^0(X, \mathcal{O}_X(D))$, $s \neq 0$ which is unique up to scalar multiplication, such that $\mathrm{Div}(s) = D$. The map $D \mapsto \mathcal{O}_X(D)$ identifies $\mathrm{Pic}(X)$ with the group of linear equivalence classes of divisors on X. (see [Bea96], I.1) (Do we need smooth X?)(!?).

Proposition 2.8.2 ([EH16], Proposition 1.30). *If X is a variety of dimension n, then c_1 is a group homomorphism*

$$c_1 \colon \mathrm{Pic}(X) \to A^1(X).$$

If X is smooth, then c_1 is an isomorphism.

[EH16] If $Y \subseteq X$ is a divisor on a smooth variety X, then the ideal sheaf of Y is a line bundle denoted $\mathcal{O}_X(-Y)$, and its inverse in the Picard group is denoted $\mathcal{O}_X(Y)$. The inverse of the map c_1 takes $[Y]$ to $\mathcal{O}_X(Y)$.

2.8.2 Chern classes of vector bundles

We now extend the definition of Chern classes to vector bundles of higher rank. Let $\mathcal{E}$ be a vector bundle on an n-dimensional variety X, and let $r = \mathrm{rank}(\mathcal{E})$. We follow the approach of [EH16] section 5.2, giving an informal characterization rather than a formal definition.

In the case $r = 1$, the Chern class $c_1(\mathcal{L})$ can be viewed as a measure of non-triviality. If $c_1(\mathcal{L}) = 0$, then $\mathcal{L}$ has a nowhere-vanishing section, and $\mathcal{L} \cong \mathcal{O}_X$. We extend this idea to higher rank vector bundles by considering the "degeneracy locus" of collections of sections. Roughly, this is the locus where the sections become linearly dependent in the fibers of $\mathcal{E}$.

More precisely, the bundle $\mathcal{E}$ is said to be trivial if and only if it has r global sections which are everywhere linearly independent. In this case, any set of r general sections will do. We look at the exterior powers of $\mathcal{E}$. The $(r-i)$-th exterior power $\bigwedge^k \mathcal{E}$ is a vector bundle of rank $\binom{r}{r-i}$, and general sections $s_1, \dots, s_{r-i}$ induce sections of $\bigwedge^{r-i} \mathcal{E}$ given by $s_1 \wedge s_2 \wedge \cdots \wedge s_{r-i}$. For $i = 0$, we have a single section $s_1 \wedge s_2 \wedge \cdots \wedge s_r$ of the line bundle $\bigwedge^r \mathcal{E}$, and the zero locus is by definition $c_1(\bigwedge^r \mathcal{E})$. This is a class in $A^1(X)$, and we call it the *first Chern class of* $\mathcal{E}$.

Necessary?

In general, we have that for $(r-i)$ general sections, we can consider the vanishing locus of

$$s_1 \wedge \cdots \wedge s_{r-i} \in \bigwedge^{r-i} \mathcal{E}.$$

This vanishing locus is called the *i-th degeneracy locus* of the sections $s_1, \dots, s_{r-i}$.

2.8.3 Important results on Chern classes

Theorem 2.8.3 ([EH16], Theorem 5.3 (a), (c)). *There is a unique way of assigning to each vector bundle $\mathcal{E}$ on a smooth quasi-projective variety X a class $c(\mathcal{E}) = 1 + c_1(\mathcal{E}) + \ldots + c_n(\mathcal{E}) \in A(X)$ in such a way that*

- *If $\mathcal{L}$ is a line bundle, then $c(\mathcal{L})$ is $1 + c_1(\mathcal{L})$, where $c_1(\mathcal{L}) \in A^1(X)$ is the class of the divisor of zeros minus the divisor of poles of any rational section of $\mathcal{L}$.*
- *If $0 \to \mathcal{E} \to \mathcal{F} \to \mathcal{G} \to 0$ is an exact sequence of vector bundles on X, then $c(\mathcal{E}) = c(\mathcal{F})c(\mathcal{G}) \in A(X)$.*

Theorem 2.8.4 ([EH16], Theorem 5.12, Splitting principle). *Any identity among Chern classes of bundles that is true for bundles that are direct sums of line bundles is true in general.*

Corollary 2.8.5 ([EH16], Corollary 5.5). *If $\mathcal{E}$ is the direct sum of line bundles $\mathcal{L}_i$, then*

$$c(\mathcal{E}) = \prod c(\mathcal{L}_i) = \prod (1 + c_1(\mathcal{L}_i)).$$

Se veiledning fra 14.03.

Corollary 2.8.6 ([EH16], Example 5.13). *If $\mathcal{E}$ is a rank r vector bundle, then $c_i(\mathcal{E}) = 0$ for $i > r$.*

Examples:

- Duals.
- Symmetric squares.
- Tensor products.

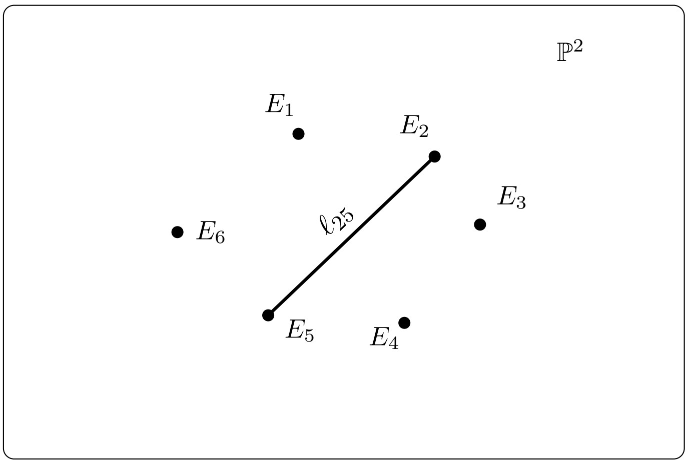

Figure 2.1: The projective plane $\mathbb{P}^2$ with six marked blow-up points and a strict transform of the line ℓ_{25}.

fig:blowup6

2.9 Lines on a cubic surface

From Corollary 2.6.6, we know that a generic smooth cubic surface contains finitely many lines. We will now show that it contains exactly 27 lines. This is a very famous result, and it was first proved in 1849 by Arthur Cayley and George Salmon, in a proof we will give a sketch of(!?).

Consider continuing this historical view.

Theorem 2.9.1. *Each smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines.*

This will be the main theorem of this section. We will prove it in two different ways. The first will be a well known proof, relating cubic surfaces to blow-ups of $\mathbb{P}^2$ in 6 points. The second proof will use intersection theory on the Grassmannian $\mathbb{G}(1,3)$.

Describe second result better.

We begin with the first proof.

2.9.1 Cubic surfaces as blow-ups of $\mathbb{P}^2$ in 6 points

Make a better figure of the blow-up of $\mathbb{P}^2$ in 6 points.

This proof will rely on theory which is not in the scope of this thesis, but it will allow us to draw a picture to show a representation of the 27 lines themselves, which we will use later.

The points $P_1, \ldots, P_6$ in $\mathbb{P}^2$ are said to be in *general position* if no three of the points are collinear, and no six lie on a conic.

rop:blowup6 Proposition 2.9.2. *([Har77], 4.7/4.7.3) Let $P_1, \ldots, P_6 \in \mathbb{P}^2$ be in general position. Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X is isomorphic to the blow-up of $\mathbb{P}^2$ in the points $P_1, \ldots, P_6$.*

This is part of a bigger class of surfaces, those being the *Del Pezzo* surfaces. These are surfaces which (Consider continuing, and maybe cite Beauville if so)(!?)

prop:27 Proposition 2.9.3. *([Har77], 4.9) Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X contains exactly 27 distinct lines. Each having self-intersection -1, and they are the only irreducible curves with negative self-intersection. They are:*

- *The exceptional curves $E_1, \dots E_6$ (6 of these)*
- *The strict transform of lines in $\mathbb{P}^2$ containing pairs of points P_i, P_j, $i \neq j$ (15 of these).*
- *The strict transform of conics in $\mathbb{P}^2$ not containing P_i (6 of these)*

2.9.2 Cayley-Salmon (!?)

Decide if we want to include this classic proof. We could do as EO, Finding 27 lines on the Fermat cubic.

2.9.3 Proof by degree of Chern classes

Theorem 2.9.4 ([EH16], Theorem 5.1). *Each smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines.*

Proof. Given a smooth cubic surface $X \subset \mathbb{P}^3$ determined by the vanishing of a cubic form F in four variables, we wish to determine the degree of the locus in $\mathbb{G}(1,3)$ of lines contained in X.

Observe that if we fix a line L in $\mathbb{P}^3$, then the condition that L lies in X is equivalent to the condition that the restriction of F to L is identically zero. Note that the restriction map from the 20-dimensional vector space of cubic forms on $\mathbb{P}^3$ to the 4-dimensional vector space of cubic forms on L, $V_L = H^0(L, \mathcal{O}_L(3)$ is a linear surjection, and the condition $L \subseteq X$ is that F maps to 0 in V_L. This imposes four linear conditions on the coefficients of F(!?).

As the line varies over $\mathbb{G}(1,3)$, the vector spaces V_L fit together to form a vector bundle of rank 4, $\mathcal{V}$ on $\mathbb{G}(1,3)$, with fiber over the point $[L]$ being V_L. A cubic form F on $\mathbb{P}^3$, through its restriction to each V_L, defines a global section σ_F of this vector bundle. Thus, the locus of lines contained in the cubic surface X is the zero locus of the section σ_F. Assuming that this zero locus is zero-dimensional(!?), its class in $A(\mathbb{G}(1,3))$ is given by the top Chern class $c_4(\mathcal{V})$ of the vector bundle $\mathcal{V}$. The number of lines contained in X is therefore the degree of this class(!?).

To compute this degree, we first express $\mathcal{V}$ in terms of the universal sub-bundle $\mathcal{S}$ on $\mathbb{G}(1,3)$. Recall that the fiber of $\mathcal{S}$ over a point $[L]$ is the 2-dimensional vector space corresponding to the line $L \subseteq \mathbb{P}^3$. The restriction of cubic forms on $\mathbb{P}^3$ to L factors through the symmetric third power of this vector space, so we have an isomorphism of vector bundles

$$\mathcal{V} \cong \mathrm{Sym}^3(\mathcal{S}^\vee).$$

Using the splitting principle, we can compute the Chern classes of $\mathcal{V}$ in terms of the Chern classes of $\mathcal{S}^\vee$. Let the Chern roots of $\mathcal{S}^\vee$ be α_1 and α_2. Then the Chern roots of $\mathcal{V}$ are given by

$$3\alpha_1, 2\alpha_1 + \alpha_2, \alpha_1 + 2\alpha_2, 3\alpha_2.$$

Thus, the top Chern class of $\mathcal{V}$ is

$$\begin{aligned} c_4(\mathcal{V}) &= (3\alpha_1)(2\alpha_1 + \alpha_2)(\alpha_1 + 2\alpha_2)(3\alpha_2) \\ &= 9\alpha_1\alpha_2(2\alpha_1^2 + 5\alpha_1\alpha_2 + 2\alpha_2^2) \\ &= 9c_2(\mathcal{S}^\vee)(2c_1(\mathcal{S}^\vee)^2 + 5c_2(\mathcal{S}^\vee)). \end{aligned}$$

To compute the degree of this class, we express it in terms of the Schubert classes $\sigma_1 = c_1(\mathcal{S}^\vee)$ and $\sigma_{1,1} = c_2(\mathcal{S}^\vee)$, and use the intersection numbers computed in (!?). Therefore, a smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines. $\square$

Finish proof properly.

2.10 Eckardt points

Introduce Eckardt points, how many there are in cubic surface in $\mathbb{P}^3$, and the pencil example. Decide if this is viable for the thesis.

Chapter 3

Results

results

3.1 Surfaces including intersecting lines

Align name of lines better

In this section we will study the problem of finding a smooth hypersurface $X_d \subset \mathbb{P}^3$ of degree d, containing an m-*cross*.

Definition 3.1.1. An m-cross is a union of m lines $l_1, l_2, \ldots, l_m$ in $\mathbb{P}^3$ which intersect at a common point p and lying in a common plane H. We will write $Z_m = \bigcup_{i=1}^m l_i$.

We require the lines to lie in a plane because if this is not true, then the point p gives rise to a tangent space of dimension 3, which cannot be included in a surface in $\mathbb{P}^3$.

Make this argument rigorous, maybe a proposition.

It is worth noting the specific question we are asking, which is that given an m-cross, does there exist a smooth hypersurface of degree d including it. We do not ask whether an m-cross is to be expected inside a general hypersurface.

Remark 3.1.2. Note that for automorphisms of $\mathbb{P}^3$, lines are sent to lines. Therefore, we can study the lines $l_1 = (\alpha : \beta : 0 : 0),\ l_2 = (\alpha : 0 : \gamma : 0),\ l_3 = (\alpha : \delta : \lambda_3\delta : 0), \ldots, l_m = (\alpha : \delta : \lambda_m\delta : 0)$ for $\lambda_i \in \mathbb{N}$ by taking the automorphism of $\mathbb{P}^3$ which sends these lines to the given lines, and then study the hypersurface of degree d in $\mathbb{P}^3$ which contains these lines. This automorphism will also take a smooth hypersurface of degree d to a smooth

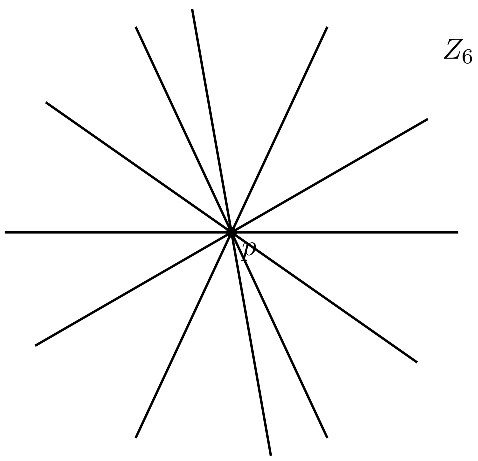

fig:Z_6

Figure 3.1: An example of a 6-cross in $\mathbb{P}^3$.

hypersurface of degree d, so showing properties for lines in this form is sufficient for the general case.

We are interested in both the existence and non-existence of degree d hypersurfaces containing m-crosses. We first consider surfaces with degree less than k, but more than 1 (degree 1 provides us a simple choice of surface including an m-cross, namely H).

lem:H

Lemma 3.1.3. *If S is a surface of degree $d > 1$, it does not contain a plane.*

Proof. Assume for contradiction that $S = V(g)$ contains a plane H. We let $H = V(x_3)$ without loss of generality (refer to a lemma or previous discussion?)(!?). Then x_3 divides g, and we write $g = x_3 h$ for some $h \in \Bbbk[x_0, x_1, x_2, x_3]_{d-1}$. We then use the Jacobi criterion (!?) on $x_3 h$, and get that S is singular on the whole of $V(x_3) \cap V(h)$. This is non-empty by Bezout's theorem (Theorem 1.1.2), giving a contradiction. (Use $d > 1$ to see that $\deg(h) = d - 1 > 0$)(!?). □

Lemma 3.1.4. *Given an m-cross, there does not exist any smooth hypersurface of degree $1 < d < m$ such that the lines are contained in the hypersurface.*

Proof. Assume for contradiction that there exists a smooth hypersurface of degree $1 < d < m$ such that the lines are contained in the hypersurface. Then we can fix another line ℓ in H, which does not intersect p, and which does not lie in the hypersurface X_d by Lemma 3.1.3. ℓ will intersect each of the lines at a point p_i, all being distinct since the lines only intersects in p. ℓ will also intersect X_d at the points p_i, thus intersecting the hypersurface at m points. By Bezout's theorem (Theorem 1.1.2), the intersection of a line with a hypersurface of degree d is at most d points, giving a contradiction.

Can't the line stay on our surface when going from one line to the other?

□

We now move on to the case of existence, with the following proposition:

Proposition 3.1.5. *For a given m-cross, we can find a smooth hypersurface of degree m such that the lines are contained in the hypersurface.*

Proof. Let $\mathfrak{L}$ be the linear system of hypersurfaces of degree m in $\mathbb{P}^3$ containing the lines $l_1, \dots, l_k$ (Prove this is not empty, because of singular surfaces.)(!?). Recall that Bertini's theorem (Theorem 1.1.1) tells us that the general member of $\mathfrak{L}$ is smooth outside the base locus of $\mathfrak{L}$. We can therefore narrow our search of singularities to the base locus.

prove

The base locus of $\mathfrak{L}$ is the union of the lines $l_1, \dots, l_k$. Thus, we must find a function $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_m$ such that $V(f)$ is smooth on the lines $l_1, \dots, l_m$. Bertini's theorem then gives us that we can find a hypersurface which is also smooth outside the base locus of $\mathfrak{L}$.

We study the function $f = g(x_1, x_2) + h(x_0, x_3)$, where $h = x_0^{m-1} x_3$ and $g = \prod l_i = x_1 x_2 \prod_{i=3}^{k} (x_1 - \lambda_i x_2)$, giving the following polynomial:

$$f(x_0, x_1, x_2, x_3) = x_1 x_2 \prod_{i=3}^{m} (x_1 - \lambda_i x_2) + x_0^{m-1} x_3$$

We can see that f is a polynomial of degree m, and that f includes the lines $l_1, \dots, l_m$, since $f(\alpha : \beta : 0 : 0) = f(\alpha : 0 : \gamma : 0) = f(\alpha : \delta : \lambda_i \delta : 0) = 0$. We must then check that f is smooth on the base locus of $\mathfrak{L}$, which is to check that $\bigcap_{i=0}^{3} (\partial_i f = 0) = \emptyset$.

We get the following derivatives:

$$\partial_0 f = \partial_0 g + \partial_0 h = 0 + (m-1)x_0^{m-2}x_3 = (m-1)x_0^{m-2}x_3$$
$$\partial_1 f = \partial_1 g + \partial_1 h = x_2 \prod_{i=3}^{m}(x_1 - \lambda_i x_2) + x_1 x_2 \sum_{i=3}^{m} \prod_{\substack{j=3 \\ j\neq i}}^{m} (x_1 - \lambda_j x_2)$$
$$\partial_2 f = \partial_2 g + \partial_2 h = x_1 \prod_{i=3}^{m}(x_1 - \lambda_i x_2) + x_1 x_2 \sum_{i=3}^{m} (-\lambda_i) \cdot \prod_{\substack{j=3 \\ j\neq i}}^{m} (x_1 - \lambda_j x_2)$$
$$\partial_3 f = \partial_3 g + \partial_3 h = 0 + x_0^{m-1} = x_0^{m-1}$$

We then find that

- $\partial_0 f = 0$ on the entire base locus.
- $\partial_1 f = 0$ on the line $l_0 = (\alpha : \beta : 0 : 0)$.
- $\partial_2 f = 0$ on the line $l_1 = (\alpha : 0 : \gamma : 0)$.
- $\partial_3 f = 0$ if and only if $x_0 = 0$, which is the case for the following points: $(0 : \beta : 0 : 0), (0 : 0 : \gamma : 0), (0 : \delta : \lambda_i \delta : 0)$ for $3 \leq i \leq m$.

We can see that the intersection of all of these zero loci is empty, showing that the function f is smooth on the base locus of $\mathfrak{L}$. Thus, we can find a smooth hypersurface of degree k such that the lines are contained in the hypersurface.

□

By the results above, we get the following corollary:

Corollary 3.1.6. *Given an m-cross, we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface if and only if $d \geq m$ or $d = 1$.*

Proof. If we have m lines, we can find a smooth hypersurface of degree $d \geq m+1$ by including arbitrary lines $l_{m+1}, l_{m+2}, \dots, l_d$ lying in the plane. For the case $d = 1$, we can take the plane H containing the lines. □

3.2 The set of degree d surfaces including m-crosses

In the previous section we studied the existence and non-existence of smooth hypersurfaces of degree d including an m-cross. We will now study the dimension of the set of degree d surfaces including m-crosses. Let Z_m be an m-cross lying in a plane H, which we will assume is the plane $V(x_3)$ by an automorphism of $\mathbb{P}^3$.

Proposition 3.2.1. *Let $A_m = \{S_d \subseteq \mathbb{P}^3 \mid Z_m \subset S_d\} \subseteq \{S_d \subseteq \mathbb{P}^3\}$ be the set of degree d surfaces containing a general Z_m. Then* $\dim A_m = \binom{d-m+2}{2} + \binom{d+2}{3} - 1$.

Proof. Let $S_d = V(f)$ for some $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_d$ be a surface which includes Z_m. Since Z_m is included in H, we write f as $f = g(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$, separating the part of f, $x_3 h(x_0, x_1, x_2, x_3)$, which vanishes on the plane. Saying that $Z_m \subseteq S_d$ is the same as saying $f|_{Z_m} = 0$, which implies that $g(x_0, x_1, x_2)|_{Z_m} = 0$, since $x_3 h(x_0, x_1, x_2, x_3)|_{Z_m}$ already vanishes. This implies that $l_1 l_2 \cdots l_m$ divides $g(x_0, x_1, x_2)$, meaning $g(x_0, x_1, x_2) = l_1 l_2 \dots l_m \cdot j(x_0, x_1, x_2)$ for some $j(x_0, x_1, x_2) \in \mathbb{C}[x_0, x_1, x_2]_{d-m}$.

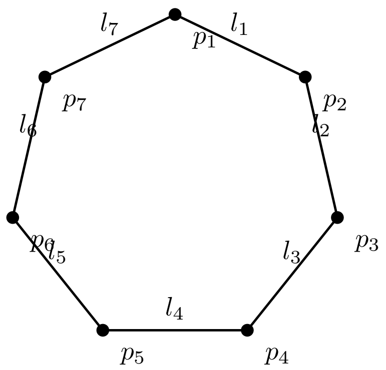

fig:m-gon

Figure 3.2: 7-gon drawn in a plane; each vertex is an intersection point p_i and the line segments represent the lines l_i.

Thus, we can write f as $f = l_1 l_2 \cdots l_m \cdot j(x_0, x_1, x_2) + x_3 h(x_0, x_1, x_2, x_3)$, where $h(x_0, x_1, x_2, x_3) \in \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1}$.

Since we can choose $j(x_0, x_1, x_2)$ and $h(x_0, x_1, x_2, x_3)$ freely, we get that $\dim A_m = \dim \mathbb{C}[x_0, x_1, x_2]_{d-m} + \dim \mathbb{C}[x_0, x_1, x_2, x_3]_{d-1} = \binom{d-m+2}{2} + \binom{d+2}{3} - 1$.

Should we subtract 1 to account for it being a projective space?

□

Include examples?

Continue with the incidence correspondence?

3.3 Surfaces including m-gons

In this section, we will consider m-gons in $\mathbb{P}^3$ and their relationship with hypersurfaces.

Definition 3.3.1. An m-gon is a union of m lines $l_1, l_2, \ldots, l_m$ such that l_i intersects l_{i+1} for $1 \leq i < m$, and l_m intersects l_1. We will write $C_m = \bigcup_{i=1}^m l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \leq i < m$, and $p_m = l_m \cap l_1$. A *regular* m-gon is an m-gon such that $l_i \neq l_j$ for $i \neq j$, and l_i intersects only the other lines as described above.

Definition 3.3.2. A *regular* m-gon is an m-gon such that $l_i \neq l_j$ for $i \neq j$, and no lines other than the adjacent ones intersect.

The notion of a regular m-gon is important. For example, a non-regular m-gon includes m lines all intersecting at a single point, i.e. an m-cross. Such cases would not hold the same properties as what one would expect from m-gons such as the one drawn in Fig. 3.2.

Definition 3.3.3. Let $\mathcal{C}_m$ be the family of all m-gons in $\mathbb{P}^3$.

Lemma 3.3.4. *Dimension and irreducibility.*

Lemma 3.3.5. *The set of regular m-gons is an open subset of $\mathcal{C}_m$.*

Fill in.

Make lines go past intersection points in figure.

prop:m-gon

Proposition 3.3.6. *Let C_m be an m-gon in $\mathbb{P}^3$. Then we can find a hypersurface of degree d such that $C_m \subseteq S_d$ when $m \leq \frac{(d+3)(d+2)(d+1)}{6d}$.*

Proof. We can look at global sections of twists of the ideal sheaf $\mathcal{I}_{C_m}$ of the m-gon to see if the m-gon can be contained in a hypersurface. We have the short exact sequence Why?

$$0 \to \mathcal{I}_{C_m} \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_{C_m} \to 0.$$

We twist this by $\mathcal{O}_{\mathbb{P}^3}(d)$ to get

$$0 \to \mathcal{I}_{C_m}(d) \to \mathcal{O}_{\mathbb{P}^3}(d) \to \mathcal{O}_{C_m}(d) \to 0.$$

From this we get the long exact sequence in cohomology:

$$\begin{tikzcd}
0 \arrow[r] & H^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \arrow[r] & H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^0(C_m, \mathcal{O}_{C_m}(d)) \arrow[dll] \\
& H^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \arrow[r] & H^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^1(C_m, \mathcal{O}_{C_m}(d)) \arrow[dll] \\
& H^2(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \arrow[r] & H^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & \dots
\end{tikzcd}$$

From [EO25] Theorem 18.23, we know that $H^i(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) = 0$ for $i > 0$ and $d \geq -3$. This means we can focus on the first part of the sequence:

$$0 \to H^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \to H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \to H^0(C_m, \mathcal{O}_{C_m}(d)) \to H^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) \to 0.$$

Since taking dimensions in cohomology is additive, we get

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) + h^1(\mathbb{P}^3, \mathcal{I}_{C_m}(d)) - h^0(C_m, \mathcal{O}_{C_m}(d)).$$

By Theorem 18.4 (!?) in [EO25] we get that $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = \binom{d+3}{3}$. $h^0(C_m, \mathcal{O}_{C_m}(d))$ can be computed by considering the short exact sequence:

$$0 \to \mathcal{O}_{C_m}(d) \to \bigoplus_{i=1}^{m} \mathcal{O}_{l_i}(d) \to \bigoplus_{i=1}^{n} \mathcal{O}_{p_i} \to 0.$$

This gives us $h^0(C_m, \mathcal{O}_{C_m}) = m \cdot \binom{d+1}{1} - m = md$.

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_m}) \geq \binom{d+3}{3} - md.$$

Meaning we can guarantee a degree d surface when $\binom{d+3}{3} - md \geq 0$, or equivalently when $m \leq \frac{(d+3)(d+2)(d+1)}{6d}$. □

Make proof more coherent, we jump around a bit.

To get an exact value for $h^0(\mathbb{P}^3, \mathcal{I}_{C_m})$, we would need to compute $h^1(\mathbb{P}^3, \mathcal{I}_{C_m})$. This issue will be explored further in Section 3.5.

Explain significance of h^0, and do it before this proof.

Remark 3.3.7. Note that this argument does not guarantee that the surface is smooth, only that it exists.

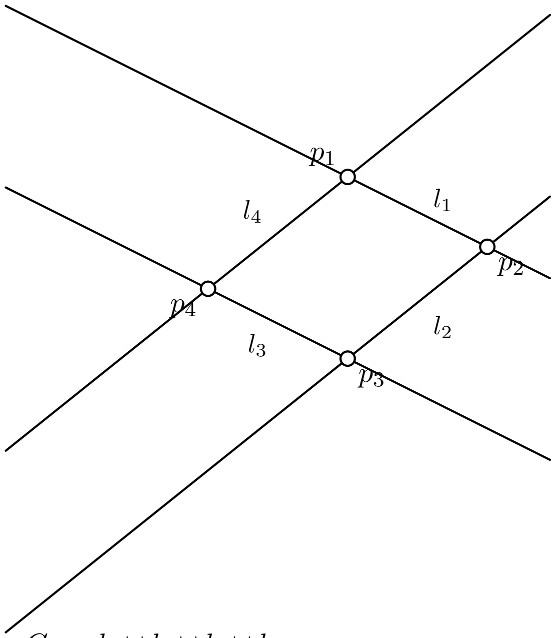

fig:4-gon

Figure 3.3: A 4-gon in $\mathbb{P}^3$, drawn on a plane.

3.4 Computational results

We should probably move the code to the appendix, and just discuss the results here.

We have done computational experiments in Macaulay2 to get a better understanding of the inequality in Proposition 3.3.6. We have tested the existence and smoothness of hypersurfaces including m-gons for various degrees d and number of lines m. The code used for these experiments is included in Section A.1, where we also discuss the details of the code. The results are given below.

Improve code explanation, and include results. Focus on main ideas of the code?

3.5 Cohomology of a 4-gon

sec:cohomology

As mentioned earlier, to get an exact value for $h^0(\mathbb{P}^3, \mathcal{I}_{C_m})$, we want to compute $h^1(\mathbb{P}^3, \mathcal{I}_{C_m})$. We will now explore this issue for the case of a 4-gon and looking at a suitable resolution of the ideal sheaf of the 4-gon. First we claim that the intersection points of a 4-gon can, by an automorphism of $\mathbb{P}^3$, be expressed as the points $(1:0:0:0), (0:1:0:0), (0:0:1:0), (0:0:0:1)$. Then the 4-gon can be expressed as the union of the lines $l_1 = V(x_2, x_3), l_2 = V(x_0, x_3), l_3 = V(x_0, x_1), l_4 = V(x_1, x_2)$. Now the ideal of the 4-gon is given by $I_{C_4} = (x_2, x_3) \cap (x_0, x_3) \cap (x_0, x_1) \cap (x_1, x_2) = (x_1x_3, x_0x_2)$. We justify our chose of points in the following lemma.

Lemma 3.5.1. *Let C_4 be a 4-gon in $\mathbb{P}^3$. Then there exists an automorphism of $\mathbb{P}^3$ such that the intersection points of C_4 are given by the points* $(1:0:0:0), (0:1:0:0), (0:0:1:0), (0:0:0:1)$.

Proof. Let p_1, p_2, p_3, p_4 be the intersection points of C_4. To be in general position in $\mathbb{P}^3$, we require that no three of the points are collinear, and that the points do not all lie in a plane. We can then find a projective transformation sending p_1 to $(1:0:0:0)$, p_2 to $(0:1:0:0)$, and p_3 to $(0:0:1:0)$. Since no three of the points are collinear, p_4 cannot be sent to a point on the line between any two of the other points. Furthermore, since the points do not all lie in a plane, p_4 cannot be sent to a point in the plane spanned by any three of the other points. Then we can send p_4 to $(0:0:0:1)$, and we are finished. □

We will now look at a resolution of the ideal sheaf of the 4-gon. From the discussion above, we have that $I_{C_4} = (x_1x_3, x_0x_2)$. This ideal has the following free resolution:

$$0 \to \mathcal{O}_{\mathbb{P}^3}(-4) \xrightarrow{\begin{pmatrix} -x_0x_2 \\ x_1x_3 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4} \to 0.$$

Finish argument.

3.6 Surfaces including m-chains

Definition 3.6.1. An m-chain is a union of m lines $l_1, l_2, \ldots, l_m$ such that l_i intersects l_{i+1} for $1 \le i < m$. We will write $C_m = \bigcup_{i=1}^m l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \le i < m$. A *regular* m-chain is an m-chain such that no line l_i intersects any l_j for $|i-j| > 1$, and $l_i \neq l_j$ for $i \neq j$.

Improve regular definition wording.

Notice that an m-chain is an $(m+1)$-gon excluding the line l_m intersecting l_1. One can then modify the code above ("as in" cite appendix)(!?) to create m-chains instead of $(m+1)$-gons, by excluding the final pair of points. The results from this are similar to those of m-gons, in that for higher degrees, we find smooth surfaces, up until we find no surfaces. There are however interesting exceptions for degree 2 and 3 surfaces. Here we find singular surfaces when considering 4-chains and 6-chains respectively. We will handle the cases separately. First, we will study the set of m-chains in $\mathbb{P}^3$. We will mostly use this in the case of 6-chains, but the results hold in general. (Do we want to use this for 4-chains also?)(!?)

Include m2 results for m-chains

Definition 3.6.2. We define $Q_m \subseteq \mathbb{G}(1,3)^m$ to be the subset of m-tuples of lines $(l_1, \ldots, l_m)$ such that they form an m-chain.

lem:dimQm Lemma 3.6.3. *Q_m is an irreducible variety of dimension $3m+1$ for $m \ge 1$ (Do we have to require $m \ge 1$, or is it obvious!?).*

Irreducibility is discussed in EH 3.3.1, saying Schubert cells and their closures are irreducible.

Proof. We observe that Q_m can be constructed as a product of Grassmannians, with a condition of intersection added for $m \ge 2$. That is, if we are given a line $L \in \mathbb{G}(1,3)$, we can consider the space of lines intersecting L, denoted as $\Sigma_L = \{l \in \mathbb{G}(1,3) \mid l \cap L \neq \emptyset\}$. Then $Q_m = \mathbb{G}(1,3) \times \Sigma_L^{m-1}$. For m $= 1$, we have that $Q_1 = \mathbb{G}(1,3)$, which is irreducible of dimension 4. Now, assume that the statement holds for $m-1$. Consider the projection map $\pi : Q_m \to Q_{m-1}$ given by $\pi(l_1, \ldots, l_m) = (l_1, \ldots, l_{m-1})$. The fiber over a point is given by $\pi^{-1}(l_1, \ldots, l_{m-1}) = \{l \in \mathbb{G}(1,3) \mid l \cap l_{m-1} \neq \emptyset\}$.

Should we denote the fiber over a point as the whole m-tuple, or just the last line? I think the first is more correct.

Note that for a given line L, Σ_L can be constructed as the image of the incidence correspondence $\Gamma = \{(l, p) \in \mathbb{G}(1,3) \times L \mid p \in l\}$ to $\mathbb{G}(1,3)$. The fiber of the projection map $f : \Gamma \to L$ is given by the set of lines through a point $p \in L$, which is isomorphic to $\mathbb{P}^2$. Then the fiber of each point in L is irreducible of dimension 2, and since f is proper, we have that Γ is irreducible of dimension 3, meaning that Σ_L is irreducible of dimension 3. Thus, Q_m is the product of $m-1$ irreducible varieties of dimension 3, and one Grassmannian of dimension 4, so Q_m is irreducible of dimension $4+3(m-1) = 3m+1$. This completes the proof. □

Update proof with notes from 09.10. Don't think this is rigorous enough to show irreducibility.

3.6.1 The case of 4-chains

Proposition 3.6.4. *For a general 4-chain in $\mathbb{P}^3$, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Cite Beauville

Proof. First, assume that S_2 is smooth. Since S_2 is a smooth quadric surface in $\mathbb{P}^3$, it is of the form $S_2 = \mathbb{P}^1 \times \mathbb{P}^1$. This means that S_2 has two rulings, meaning that there are two families of lines on S_2, such that any two lines in the same family do not intersect, and any two lines in different families intersect at exactly one point. Denote the two families as F_1 and F_2. Then we can see that a 4-chain must consist of two lines from F_1 and two lines from F_2, since any two lines in the same family do not intersect. This means that the 4-chain must be of the form l_1, l_2, l_3, l_4 where $l_1, l_3 \in F_1$ and $l_2, l_4 \in F_2$. However, this means that l_4 intersects l_1, since they are in different families, meaning that the 4-chain is in fact a 4-gon. This is a contradiction, since this is not a general 4-chain. Thus, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain. □

Proposition 3.6.5. *For a general 4-chain in $\mathbb{P}^3$, we can find a singular quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Proof. We pair the lines of the 4-chain as (l_1, l_2) and (l_3, l_4). Each pair of lines defines a plane in $\mathbb{P}^3$, denoted as $H_1 = \mathrm{span}(l_1, l_2)$ and $H_2 = \mathrm{span}(l_3, l_4)$. Since the 4-chain is general, we have that $H_1 \neq H_2$. Then the union of the two planes $S_2 = H_1 \cup H_2$ is a singular quadric surface in $\mathbb{P}^3$ that includes the 4-chain. □

Improve, used GPT here.

3.6.2 The case of 7-chains

Now we will look at the case of 7-chains in cubic surfaces.

Proposition 3.6.6. *For a general 7-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain.*

Why open?

Proof. Let $\mathcal{M}$ be the space of smooth cubic surfaces in $\mathbb{P}^3$. This is an open subset of $\mathbb{P}(\mathbb{C}[x_0, \ldots, x_3]_3) \cong \mathbb{P}^{19}$. Since $\mathcal{M}$ is an open subset, it shares the dimension, i.e. $\dim \mathcal{M} = 19$. Let $\mathcal{N} = \{(S_d, l_1, \ldots, l_7) \in \mathcal{M} \times \mathbb{G}(1,3)^7 \mid \mathcal{L} \subset S_d, \mathcal{L} \text{ is an } m\text{-chain}\}$ be the incidence variety of smooth cubic surfaces including a general 7-chain. We have the following diagram:

$$\begin{tikzcd} & \mathcal{N} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ \mathcal{M} & & \mathbb{G}(1,3)^7 \end{tikzcd} . \tag{3.1}$$

Should we use $\mathcal{Q}_7$ instead of $\mathbb{G}(1,3)^7$?

The projection map $\pi_1 : \mathcal{N} \to \mathcal{M}$ is given by $\pi_1(S_d, l_1, \ldots, l_7) = S_d$. Then the fiber over a point $S_d \in \mathcal{M}$ is given by the set of 7-chains included in S_d. Since a smooth cubic surface includes exactly 27 lines, we can find a 7-chain by choosing 7 lines from these 27. We know from Proposition 3.6.8(!?) that it is possible to find a 7-chain, and so this fiber is not empty. Furthermore, there are only a finite number of 7-chains when there is a finite number of lines. Then the dimension of the fiber $\dim \pi_1^{-1}(S_d)$ is 0, being a finite set of 7-chains. Thus, from Theorem 1.1.3, we have that $\dim \mathcal{N} = \dim \mathcal{M} + \dim \pi_1^{-1}(S_d) = 19 + 0 = 19$.

Refer to result

Result in this section

Refer to theorem about dimension of fiber.

Now we are interested in the fiber of the set of 7-chains in $\mathbb{P}^3$, $\mathcal{Q}_7 \subseteq \mathbb{G}(1,3)^7$. By Lemma 3.6.3, we have that $\dim \mathcal{Q}_7 = 22$, and so the projection map $\pi_2 : \mathcal{N} \to \mathbb{G}(1,3)^7$ given by $\pi_2(S_d, l_1, \ldots, l_7) = (l_1, \ldots, l_7)$ cannot be surjective on (dominate!?) $\mathcal{Q}_7$. This means that taking a general 7-chain $C_7 \in \mathcal{Q}_7$, and considering the fiber $\pi_2^{-1}(C_7)$, we do not expect this to be non-empty, meaning we will not have a surface including it. Thus, for a general 7-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain. □

Proof should include: projection is surjective (every smooth cubic surface has at least a 7-chain, shown in this section), implies dim N $\geq$ dim M. Then fiber is finite, so dim N $\leq$ dim M. Prop 10.26?

Again, as in the case for 4-chains in quadric surfaces, computations show that we can find a singular cubic surface including a general 7-chain.

Proposition 3.6.7. *For a general 7-chain in $\mathbb{P}^3$, we can find a* singular *cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain. (!?)*

Proof. Find a nice example of a singular cubic surface including a general 7-chain. □

Need to rework entire section above, should be 6-chains, not 7-chains.

Note that if we instead asked whether a smooth cubic surface in $\mathbb{P}^3$ can include a 7-chain, the situation is different. In this case, we can indeed find a 7-chain such that it is included in the surface. In fact, we know we could find an 11-chain in this case, and a 12-gon. We show this in the following proposition.

Maybe move this proposition earlier, to use in proof. Maybe lemma? Also interesting in itself, so maybe not.

prop:12-gon **Proposition 3.6.8.** *For a smooth cubic surface in $\mathbb{P}^3$, we can find a 12-gon such that it is included in the surface.*

Proof. Recall from Proposition 2.9.2 that a smooth cubic surface in $\mathbb{P}^3$ is isomorphic to the blow-up of $\mathbb{P}^2$ in 6 points. It may be useful to again consider the picture in figure Fig. 2.1 on page (!?). The surface includes 27 lines, which are precisely the exceptional divisors of the blow-up, the strict transforms of the 15 lines through pairs of points, and the strict transforms of the 6 conics through 5 of the points (Proposition 2.9.3). We can then find a 12-gon by considering the following lines:

Let $E_1, E_2, E_3, E_4, E_5, E_6$ be the exceptional divisors, let L_{ij} be the strict transform of the line through points p_i and p_j, and let C_i be the strict transform of the conic through all points except p_i. Then we can consider the 12-gon given by the lines $E_1, L_{12}, E_2, L_{23}, E_3, L_{34}, E_4, L_{45}, E_5, L_{56}, E_6, C_1, L_{16}$. This is indeed a (general!?) 12-gon since each line intersects exactly two other lines in the set. □

By removing one of the lines from the 12-gon, we immediately get the following:

Corollary 3.6.9. *For a smooth cubic surface in $\mathbb{P}^3$, we can find an 11-chain such that it is included in the surface.*

Fill out later.

This implies that the m-chains longer than 7 included in the smooth cubic surfaces are somehow special.

3.6.3 The case of 6-chains

Now we will examine 6-chains in cubic surfaces. We will show that a general 6-chain is not included in a smooth cubic surface, but that we can find singular cubic surfaces including a general 6-chain. We start by studying the incidence variety of smooth cubic surfaces including a general 6-chain.

Why open?

Let $\mathcal{M}$ be the set of smooth cubic surfaces in $\mathbb{P}^3$. This is an open subset of $\mathbb{P}(\mathbb{C}[x_0, \dots, x_3]_3) \cong \mathbb{P}^{19}$. Since $\mathcal{M}$ is an open subset, it shares the dimension, i.e. $\dim \mathcal{M} = 19$. Let $\mathcal{N} = \{(S_d, l_1, \dots, l_6) \in \mathcal{M} \times \mathbb{G}(1,3)^6 \mid \mathcal{L} \subset S_d, \mathcal{L} \text{ is an } m\text{-chain}\}$ be the incidence variety of smooth cubic surfaces including a general 6-chain. We have the following diagram:

$$\begin{tikzcd} & \mathcal{N} \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ \mathcal{M} & & \mathbb{G}(1,3)^6 \end{tikzcd} . \tag{3.2}$$

Should we use $\mathcal{Q}_7$ instead of $\mathbb{G}(1,3)^7$?

Remark 3.6.10. Note that it is important that we consider the incidence variety of over specifically the smooth cubic surfaces, including a general 6-chain. If not, one can notice that the fiber of the singular surfaces which are the union of 3 planes, will include a much higher dimensional family of 6-chains, and so the incidence variety will not be irreducible. (!?)

Lemma 3.6.11. *For a smooth cubic surface, we cannot find a regular(!?) 6-chain.*

Proof. We denote the 27 lines of the cubic surface as $C_1, \dots, C_27$. Let Q_5 be a regular 5-chain, consisting of $C_1, \dots, C_5$. Look at the class $D = C_1 + C_2 - C_4 - C_5$. Then $D^2 = 0$, and $HD = 0$. The Hodge index theorem(!?) tells us that $D = 0$, and we have the equality $C_1 + C_2 = C_4 + C_5$. Now we know that if a line C_i for $i \neq 4$ (then $C_4^2 + C_4 C_5 = 0$) intersects C_5, it must also intersect either C_1 or C_2, and would not make a regular 6-chain. □

3.7 Sjekk til slutt

- Skal vi skrive generic eller general? Når?
- Skrive om til m-gon/chain?
- Legg til at vi jobber med "regulære" ting.
- Endre instanser av $\mathbb{C}$ til $\Bbbk$?

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Appendix A

Programming

ec:appendix

A.1 m-gon

app:mgon
lst:m-gon

Listing A.1: m-gon

```
-- Step 1: Define the ring and parameters
kk = QQ;
RingP3 = kk[x_0..x_3];
d = 2;  -- Degree of the hypersurface
nrPoints = 4;

-- Step 2 and 3: Generate random points and create pairs of subsequent points
-- Function for generating i random points in P^3
randomPoints = i -> (
    v := for j in 0..3 list random(kk);
    -- Ensure the point is not zero
    while all(v, x -> x == 0) do (
        v = for j in 0..3 list random(kk)
    );
    v
)
-- Pick nrPoints random points in P^3
myPoints = apply(nrPoints, randomPoints);
-- Create pairs of subsequent points, and also the pair consisting of the last and first point
subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i, myPoints#(i+1)};
-- Add final pair
subsequentPairs = append(subsequentPairs, {myPoints#0, myPoints#(#myPoints-1)});

-- Step 3: Generate matrices from point pairs
-- Function to convert two points to a matrix with first row {x_0 .. x_3}
pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points#1};
-- Convert each pair of points to a matrix
pointsMatrix = apply(subsequentPairs, pointsToMatrix);

-- Step 4: Get the ideals of the lines
```

```
-- Given a matrix with first row {x_0 .. x_3} and next two rows points, return the ideal of the line through the points
lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
allLines = apply(pointsMatrix, lineFromPoints);

-- Step 5: Create the ideal of the union of all lines
-- Intersect the ideals of the lines to get ideal of the union of all lines
Iz = intersect allLines;

-- Step 6: Pick a degree d element in Iz, which defines a surface containing all the lines, and check properties
-- Pick a random degree d element in Iz
f = random(d, Iz);
-- Create the variety defined by f
Vf = variety ideal f;

-- Check if the surface is smooth and not the zero polynomial
codimension = codim singularLocus ideal f;
isZero = (f == 0)
isSingular = (codimension < 4)
isSmooth(Vf) and not isZero
```

We'll go through the code step by step.

Step 1: We begin by defining the field and the polynomial ring in four variables over this field. We then define the degree d of the surfaces we want to consider, and the number of points to use, which will be the same as the number of lines in the m-gon.

Step 2: We define a function to pick a given amount of random points in our quotient ring, ensuring they are not zero. We then apply the function to get the points, and gather the points into pairs of subsequent points so that we can later define lines between these.

Step 3: Next, we define a function for turning two points into a matrix. The function takes two points and returns a matrix, as follows:

$$((a, b, c, d), (d, e, f, g)) \mapsto \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ a & b & c & d \\ d & e & f & g \end{pmatrix}$$

Step 4: Now we can define the ideal of the line between the two points by taking the 3×3 minors of this matrix.

Step 5: Next we take the intersection of all the line ideals, to get the ideal of $Z = \bigcup l_i$,

Step 6: We choose a random member of $I_Z(d)$ and compute if this defines an empty, singular, or smooth surface.

Results: We have been interested in finding the maximum number of lines m such that we can find a smooth surface of degree d including an m-gon. The results are summarized in the following table:

Include the check that we don't use f=0 when the ideal has degree d elements.

A.2 m-chain

app:mchain
lst:m-chain

Listing A.2: m-chain

```
-- Step 1: Define the ring and parameters
```

```
kk = QQ;
-- kk = ZZ/32749;
RingP3 = kk[x_0..x_3];
d = 3;  -- Degree of the hypersurface
nrLines = 6;
nrPoints = nrLines+1;

-- Step 2 and 3: Generate random points and create pairs of subsequent points
-- Function for generating i random points in P^3
randomPoints = i -> (
    v := for j in 0..3 list random(kk);
    -- Ensure the point is not zero
    while all(v, x -> x == 0) do (
        v = for j in 0..3 list random(kk)
    );
    v
)
-- Pick nrPoints random points in P^3
myPoints = apply(nrPoints, randomPoints);
-- Create pairs of subsequent points, and also the pair consisting of the last and first point
subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i, myPoints#(i+1)};

-- Step 3: Generate matrices from point pairs
-- Function to convert two points to a matrix with first row {x_0 .. x_3}
pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points#1};
-- Convert each pair of points to a matrix
pointsMatrix = apply(subsequentPairs, pointsToMatrix);

-- Step 4: Get the ideals of the lines
-- Given a matrix with first row {x_0 .. x_3} and next two rows points, return the ideal of the line through the points
lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
allLines = apply(pointsMatrix, lineFromPoints);

-- Step 5: Create the ideal of the union of all lines
-- Intersect the ideals of the lines to get ideal of the union of all lines
Iz = intersect allLines;

-- Step 6: Pick a degree d element in Iz, which defines a surface containing all the lines, and check properties
-- Pick a random degree d element in Iz
f = random(d, Iz);
-- Create the variety defined by f
Vf = variety ideal f;

-- Check if the surface is smooth and not the zero polynomial
codimension = codim singularLocus ideal f;
isZero = (f == 0)
isSingular = (codimension < 4)
isSmooth(Vf) and not isZero
```

```

F = sheaf(Iz);
H0 = HH^0(F(d))
H1 = HH^1(F(d))
```

A.3 Code for the especially interested

One may notice that requiring the change of degree d and amount of points in the code above leads to the tedious work of changing each of these every time we calculate the program. We have used the code above especially when checking specific examples. An example is studying the ideal I_Z in more detail to see whether the degree d surface including the configurations are unique, by checking if the ideal is generated by a single degree d element. But most of the time our goal has been to see the results for several values at once, which also becomes quite time-consuming as the values get bigger. We therefore opted for a while-loop, given below. This prints results as it moves along, and has a time limit one can change if desired.

Make the code proper, and include.